

5. Sequences of functions, functionseries, elementary functions

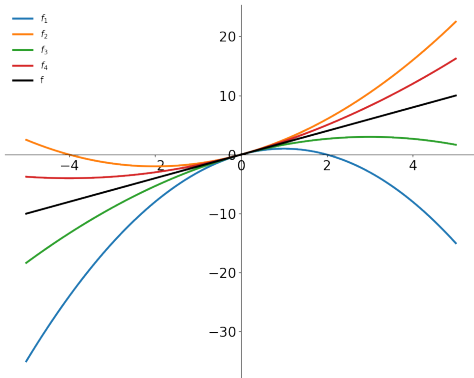
Calculus

Sequences of functions

- **Definition.** Let $D \neq \emptyset$ and

$$\mathcal{F} := \{f : D \rightarrow \mathbb{R} \mid f \text{ is a function}\}.$$

Then a map of the type $\mathbb{N} \rightarrow \mathcal{F}$ is called a sequence of functions defined on D and denoted by (f_n) , (g_n) , or (h_n) .



Pointwise convergent sequences

- Let (f_n) be a sequence of functions with domain D . Then, for any fixed $x \in D$, $(f_n(x)) \subseteq \mathbb{R}$ is a real sequence.

• **Definition (Pointwise convergence).** We say that the sequence of functions (f_n) is *pointwise convergent* on $H \subseteq D$ if the real sequence $(f_n(x))$ is convergent for all $x \in H$. Let denote D^* the largest subset of D on which (f_n) is convergent pointwise. Then the function defined by

$$F(x) := \lim_{n \rightarrow \infty} f_n(x), \quad x \in D^*$$

is called a *limit function*.

- Example.** Let $D := [0, 2]$ and $f_n : D \rightarrow \mathbb{R}$, $f_n(x) := x^n$ if $n \in \mathbb{N}$. Then

$$f_n(x) \rightarrow \begin{cases} 0 & \text{if } 0 \leq x < 1, \\ 1 & \text{if } x = 1, \\ +\infty & \text{if } 1 < x \leq 2. \end{cases}$$

Consequently, $D^* = [0, 1]$ and the limit function is

$$F(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1, \\ 1 & \text{if } x = 1. \end{cases}$$

Observe that any member of the above sequence is continuous but the limit function is not!

Uniformly convergent sequences

- **Definition (Uniform convergence).** Keeping the previous notation, we shall say that (f_n) is uniformly convergent on $H \subseteq D$ if its pointwise limit function F exists on H and, for all $\varepsilon > 0$, there exists $n_0 \in \mathbb{N}$, such that $|f_n(x) - F(x)| < \varepsilon$ holds for all $x \in H$ whenever $n \geq n_0$.
- Uniform convergence is stronger than pointwise convergence: *If a sequence of functions is uniformly convergent then it is pointwise convergent.*
- **Theorem.** If (f_n) is a uniformly convergent sequence of continuous functions, then its limit function is continuous.
- Corollary: The sequence $f_n : [0, 2] \rightarrow \mathbb{R}$, $f_n(x) := x^n$ appearing in our previous example is not uniformly convergent on $[0, 1]$.

Example of a uniformly convergent sequence

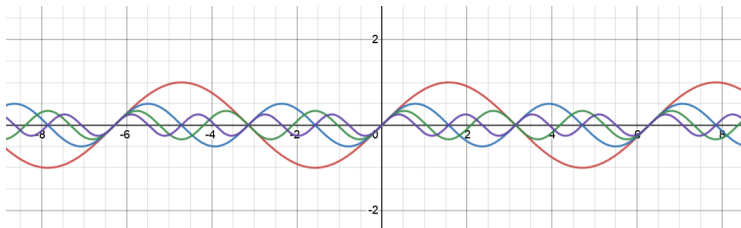
- For $n \in \mathbb{N}$, let

$$f_n : \mathbb{R} \rightarrow \mathbb{R}, \quad f_n(x) := \frac{\sin(nx)}{n}.$$

- Obviously, (f_n) is a pointwise convergent sequence on $\mathbb{R}$ with a limit function equals to zero everywhere. The convergence is uniform: Let $\varepsilon > 0$ be arbitrary. Then

$$|f_n(x) - F(x)| = \left| \frac{\sin(nx)}{n} - 0 \right| = \frac{|\sin(nx)|}{n} < \frac{1}{n}.$$

Thus let $n_0 := \lceil \frac{1}{\varepsilon} \rceil + 1$.



Function series and power series

- **Definition (Function series).** Let (f_n) be a sequence of functions. Then the sequence of functions

$$\left(\sum_{k=1}^n f_k \right)$$

is called a function series defined by (f_n) .

- *Remark: In view of the above definition, any function series is a sequence of functions. Thus it makes sense to speak about the pointwise and uniform convergence of a function series.*

- **Definition (Power series).** Let $a_0 \in \mathbb{R}$, $(a_n) \subseteq \mathbb{R}$ and $x_0 \in \mathbb{R}$. Then the function series defined by the sequence

$$f_n := \begin{cases} a_0 & \text{if } n = 0 \\ a_n(\text{id} - x_0)^n & \text{otherwise} \end{cases}$$

is called a power series. The point x_0 is called the center or the midpoint of the power series.

- *Remark: Observe that any power series is convergent in its midpoint. Thus the set of convergence of a power series cannot be empty.*

Set of convergence (Power Series)

• **Theorem (Cauchy–Hadamard).** Let (a_n) be a real sequence and $x_0 \in \mathbb{R}$. Define further the nonnegative extended real number

$$\alpha := \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}.$$

The power series

$$\sum_{n=0}^{\infty} a_n (\text{id} - x_0)^n$$

converges at a point $x \in \mathbb{R}$ if and only if

$$x \in]x_0 - r, x_0 + r[\cup \{x_0\},$$

where r is called the radius of convergence and is given by

$$r := \begin{cases} 0 & \text{if } \alpha = +\infty, \\ \frac{1}{\alpha} & \text{if } 0 < \alpha < +\infty, \\ +\infty & \text{if } \alpha = 0. \end{cases}$$

• **Remark.** Keeping the above notation, if

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

exists and finite, then its value is α .

The exponential function

- **Theorem and definition (The exponential function).** The power series

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} \quad (1)$$

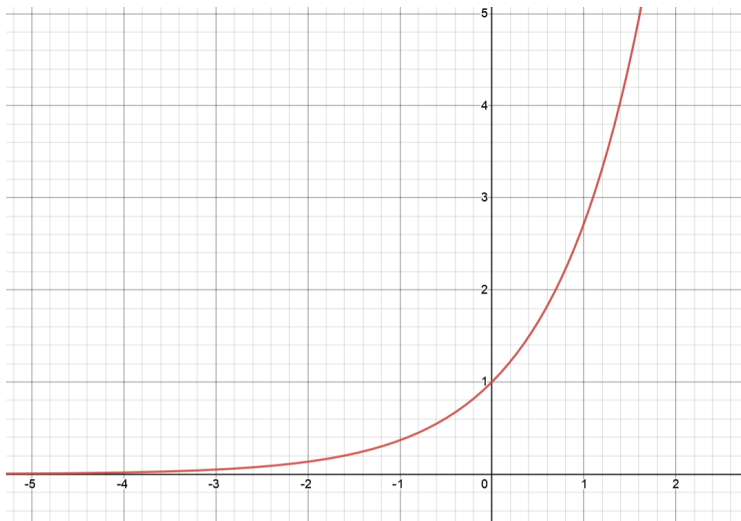
is convergent on $\mathbb{R}$. Its sumfunction is called the exponential function and denoted by

$$\exp : \mathbb{R} \rightarrow \mathbb{R}.$$

Proof (Apply Cauchy–Hadamard Theorem). Obviously, we have $a_n := \frac{1}{n!}$ and $x_0 = 0$. To prove our statement, it is enough to show that $\alpha = 0$. Indeed:

$$\alpha = \limsup_{n \rightarrow \infty} \sqrt[n]{\left| \frac{1}{n!} \right|} = \limsup_{n \rightarrow \infty} \frac{1}{\sqrt[n]{n!}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt[n]{n!}} = 0.$$

The exponential function



Properties of the exponential function

• **Theorem.** The following statements hold true.

(E1) The exponential function is *strictly increasing*, *convex*, and *continuous* on its domain. Consequently, $\exp$ is injective.

(E2) It takes only *positive* values, $\exp(0) = 1$, furthermore

$$\lim_{x \rightarrow -\infty} \exp(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow +\infty} \exp(x) = +\infty.$$

(E3) For all $x, y \in \mathbb{R}$, we have

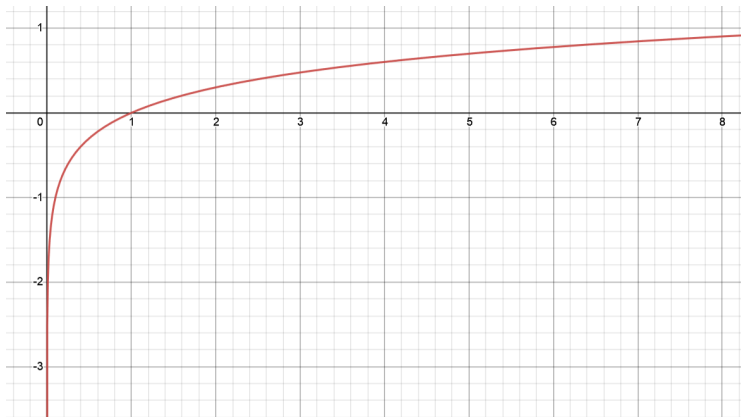
$$\exp(x + y) = \exp(x) \cdot \exp(y), \quad \text{thus } \exp(-x) = (\exp(x))^{-1}.$$

• *Remark:* In view of (E1) and (E2), the exponential function possesses an inverse function. Its inverse function is denoted by

$$\ln : \mathbb{R}_+ \rightarrow \mathbb{R}$$

and called the *logarithm function*. (Here $\mathbb{R}_+$ stands for the set of positive real numbers.)

The logarithm function



Properties of the logarithm function

• **Theorem.** The following statements hold true.

(L1) The function $\ln$ is *strictly increasing*, *concave*, and *continuous*.

(L2) It is *surjective*, $\ln(1) = 0$, furthermore

$$\lim_{x \rightarrow 0^+} \ln(x) = -\infty \quad \text{and} \quad \lim_{x \rightarrow +\infty} \ln(x) = +\infty.$$

(L3) For all $x, y \in \mathbb{R}_+$, we have

$$\ln(x \cdot y) = \ln(x) + \ln(y), \quad \text{thus} \quad -\ln(x) = \ln(x^{-1}).$$

Hyperbolic functions

• **Theorem.** The power series

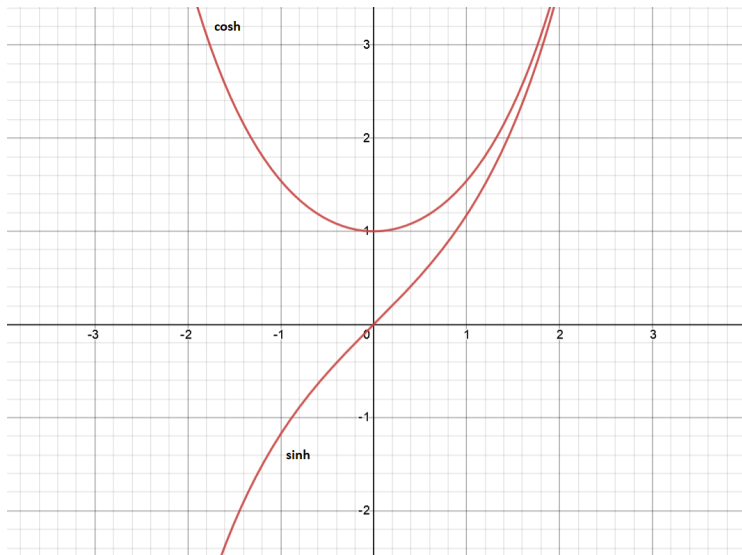
$$\sinh(x) := \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!} \quad \text{and} \quad \cosh(x) := \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

are convergent on $\mathbb{R}$.

Name of the above functions:

- ★ $\sinh$ - Hyperbolic sine,
- ★ $\cosh$ - Hyperbolic cosine.

Hyperbolic functions



Properties of hyperbolic functions

• **Theorem.** The following statements hold true.

(H1) For all $x \in \mathbb{R}$, we have

$$\sinh(x) = \frac{1}{2} (\exp(x) - \exp(-x)) \quad \text{and} \quad \cosh(x) = \frac{1}{2} (\exp(x) + \exp(-x)),$$

consequently, the hyperbolic functions are *continuous*, furthermore cosh is positive.

(H2) $\sinh(0) = 0$ and $\cosh(0) = 1$.

(H3) For all $x, y \in \mathbb{R}$,

$$\begin{aligned} \sinh(x + y) &= \sinh(x) \cosh(y) + \sinh(y) \cosh(x) \quad \text{and} \\ \cosh(x + y) &= \cosh(x) \cosh(y) + \sinh(x) \sinh(y). \end{aligned}$$

(H4) The function cosh is even, while sinh is odd.

(H5) For all $x, y \in \mathbb{R}$, we have

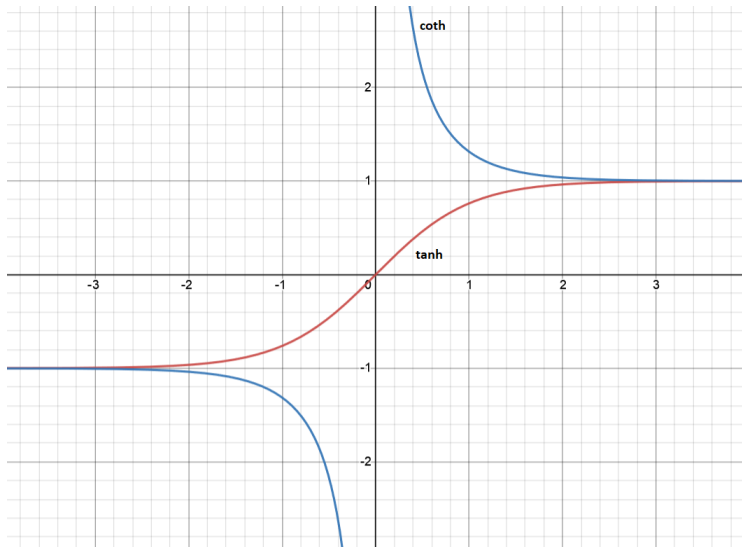
$$\cosh^2(x) - \sinh^2(x) = 1.$$

Hyperbolic tangent and cotangent

- **Definition.** The functions $\tanh : \mathbb{R} \rightarrow \mathbb{R}$ and $\coth : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ are called *hyperbolic tangent* and *hyperbolic cotangent*, respectively, and are defined by

$$\tanh(x) := \frac{\sinh(x)}{\cosh(x)} \quad \text{and} \quad \coth(x) := \frac{\cosh(x)}{\sinh(x)}.$$

Hyperbolic tangent and cotangent



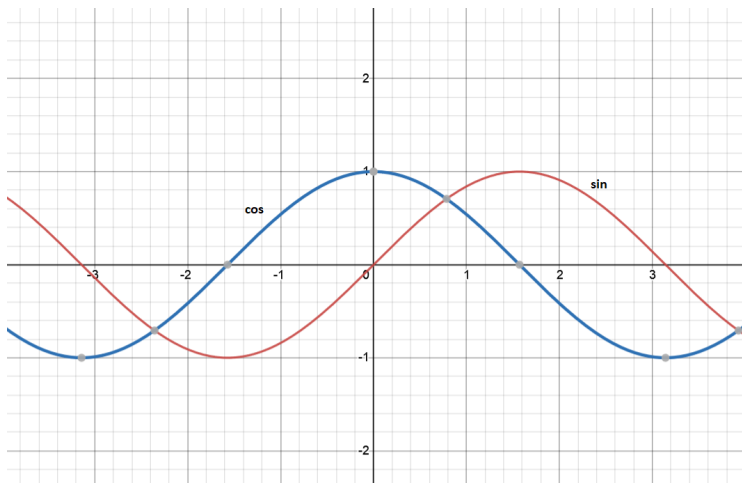
Trigonometric functions

- **Definition.** The power series

$$\sin(x) := \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \quad \text{and} \quad \cos(x) := \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

are convergent on $\mathbb{R}$ and are called *sine* and *cosine*, respectively.

Trigonometric functions



Properties of trigonometric functions

• **Theorem.** The following statements hold true.

(T1) For all $x \in \mathbb{R}$, we have

$$\sin(x) = \frac{1}{2i} (\exp(ix) - \exp(-ix)) \quad \text{and} \quad \cos(x) = \frac{1}{2} (\exp(ix) + \exp(-ix)),$$

consequently trigonometric functions are continuous.

(T2) Trigonometric functions are periodic functions with period 2π .

(T3) $\sin(0) = 0$ and $\cos(0) = 1$.

(T3) For all $x, y \in \mathbb{R}$, we have

$$\begin{aligned} \sin(x + y) &= \sin(x) \cos(y) + \sin(y) \cos(x) \quad \text{and} \\ \cos(x + y) &= \cos(x) \cos(y) - \sin(x) \sin(y). \end{aligned}$$

(T4) The function $\cos$ is even, while $\sin$ is odd.

(T5) For all $x \in \mathbb{R}$, we have

$$\cos^2(x) + \sin^2(x) = 1.$$

Tangent and cotangent

Definition. The functions

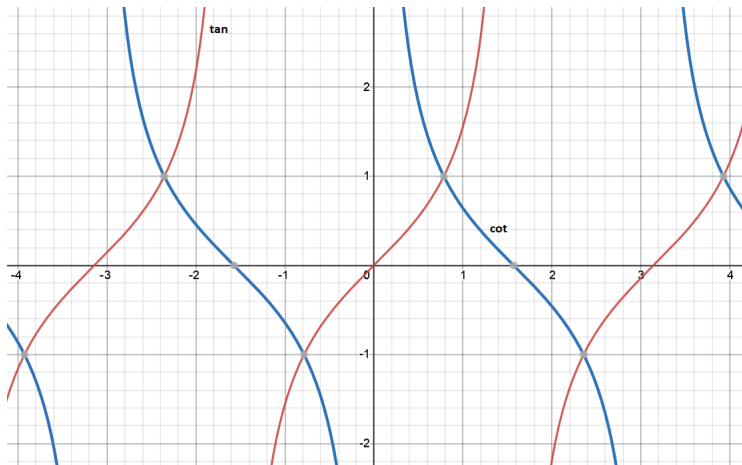
$$\tan : \{x \in \mathbb{R} \mid x \neq \frac{\pi}{2} + k\pi, \text{ if } k \in \mathbb{Z}\} \rightarrow \mathbb{R}, \quad \text{and}$$

$$\cot : \{x \in \mathbb{R} \mid x \neq k\pi, \text{ if } k \in \mathbb{Z}\} \rightarrow \mathbb{R}$$

are called *tangent* and *cotangent*, respectively, furthermore

$$\tan(x) := \frac{\sin(x)}{\cos(x)} \quad \text{and} \quad \cot(x) := \frac{\cos(x)}{\sin(x)}.$$

Tangent and cotangent



Inverse of hyperbolic functions

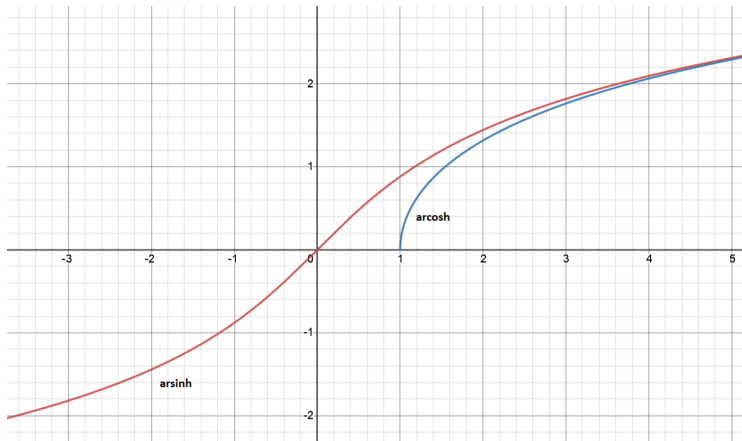
Definition. The function $\sinh$ is continuous and strictly increasing, hence it is invertible. Its inverse is called *area hyperbolic sine*. Notation: $\operatorname{arsinh} : \mathbb{R} \rightarrow \mathbb{R}$. Finally, we have

$$\operatorname{arsinh}(x) = \ln \left(x + \sqrt{x^2 + 1} \right), \quad (x \in \mathbb{R}).$$

Definition. The function $\cosh$ is continuous and strictly increasing on the set of non-negative real numbers, thus this restriction has an inverse function. This inverse is called *area hyperbolic cosine*. Notation: $\operatorname{arcosh} : [1, +\infty] \rightarrow \mathbb{R}$. Furthermore we have

$$\operatorname{arcosh}(x) = \ln \left(x + \sqrt{x^2 - 1} \right), \quad (x \in \mathbb{R} \text{ and } x \geq 1).$$

Inverse of hyperbolic functions

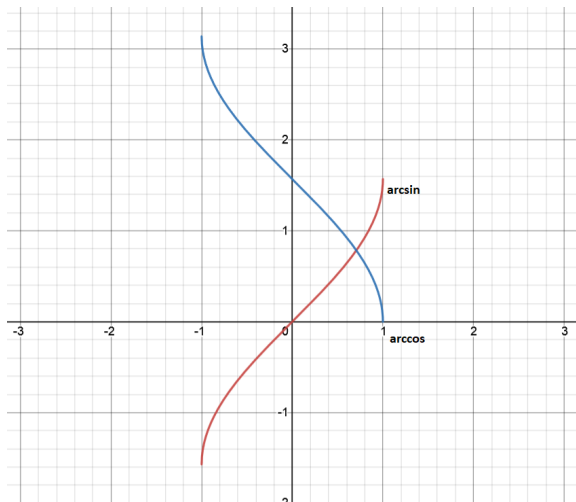


Inverse of trigonometric functions

Definition. The function $\sin$ is continuous and strictly increasing on $[-\frac{\pi}{2}, \frac{\pi}{2}]$ thus it has an inverse function on the image which is called the *arcus sine*.
Notation: $\arcsin : [-1, 1] \rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$.

Definition. The function $\cos$ is continuous and strictly decreasing on $[0, \pi]$, hence it has an inverse function on the image which is called *arcus cosine* and denoted by $\arccos : [-1, 1] \rightarrow [0, \pi]$.

Inverse of trigonometric functions



Inverse of tangent function

Definition. The function $\tan$ is continuous and strictly increasing on $[-\frac{\pi}{2}, \frac{\pi}{2}]$ hence it has an inverse on the image, which is called *arcus tangent* and denoted by $\arctan : \mathbb{R} \rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$.

Inverse of tangent function

